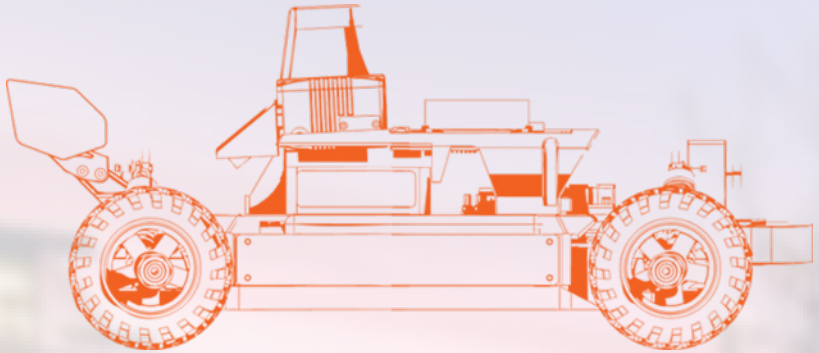


LAB-03

Reactive Autonomy



Chinmay Vilas Samak

Ph.D. Candidate

ARMLab, CU-ICAR

Email: csamak@clemson.edu



Tanmay Vilas Samak

Ph.D. Candidate

ARMLab, CU-ICAR

Email: tsamak@clemson.edu

Lab Outline and Expected Outcomes

- **Emergency braking**
  - **Formulation A:** Single measurement
  - **Formulation B:** Multiple measurements
- **Wall following**
  - **Formulation A:** Follow just one wall
  - **Formulation B:** Follow both the walls
- **Gap following**
  - **Formulation A:** Naïve gap follower
  - **Formulation B:** Disparity extender
- **GOAL:** Perception-enhanced vehicle control







Formulation A: Single Measurement

Consider a frontal laser beam a

The laser beam measures distance w.r.t. LIDAR

The LIDAR is offset from the front bumper by a distance A

Hence, the distance to collision is $DTC = a - A$

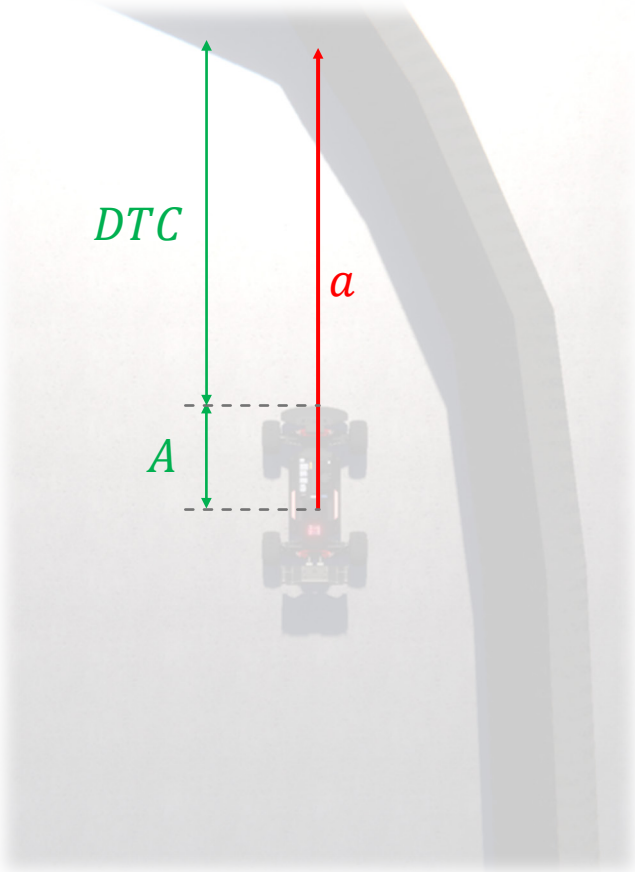
Let a safe DTC threshold be D_{th} (e.g., 3 × wheelbase)

Emergency braking behavior requires:

$$\begin{cases} DTC > D_{th}; & \text{Speed setpoint} \\ DTC \leq D_{th}; & \text{AEB triggered} \end{cases}$$

Zero steering angle is enforced: $\delta_t = 0$

Instead of hard-braking, you can implement a controller to reduce the velocity proportional to the DTC for smoother braking.



Formulation B: Multiple Measurements

Small issue: Using a single measurement can be suboptimal. A spurious return due to any reason (noise, dust, a reflection) can become the new minimum and seem like an obstacle (a.k.a. the "ghost obstacle" problem in autonomous racing). A vehicle traveling at high speeds may not have enough time to recover and may crash!

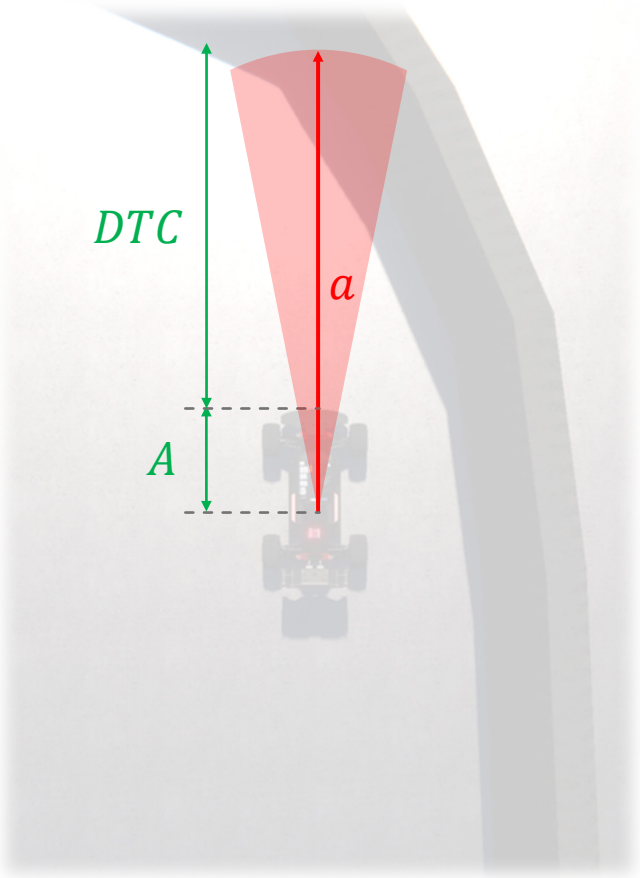
Proposed solution: We can consider multiple range measurements across a small sector on the front. The frontal distance can be then computed as the mean of these readings, which is more stable/reliable.

Let a safe DTC threshold be D_{th} (e.g., 3 × wheelbase)

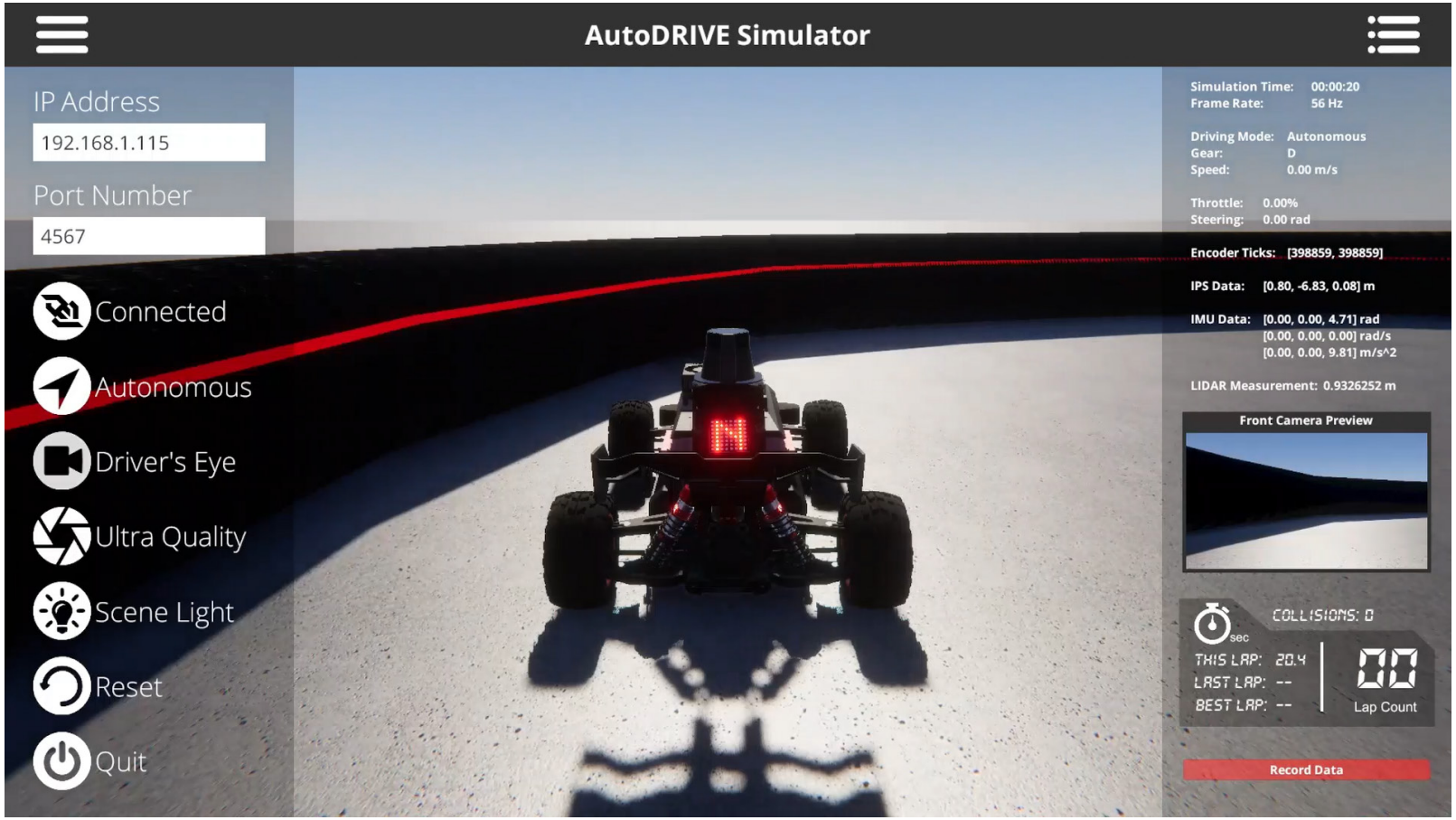
Emergency braking behavior requires:

$$\begin{cases} DTC > D_{th}; & \text{Speed setpoint} \\ DTC \leq D_{th}; & \text{AEB triggered} \end{cases}$$

Zero steering angle is enforced: $\delta_t = 0$



Emergency Braking Demo



Primary Objective:
Safety

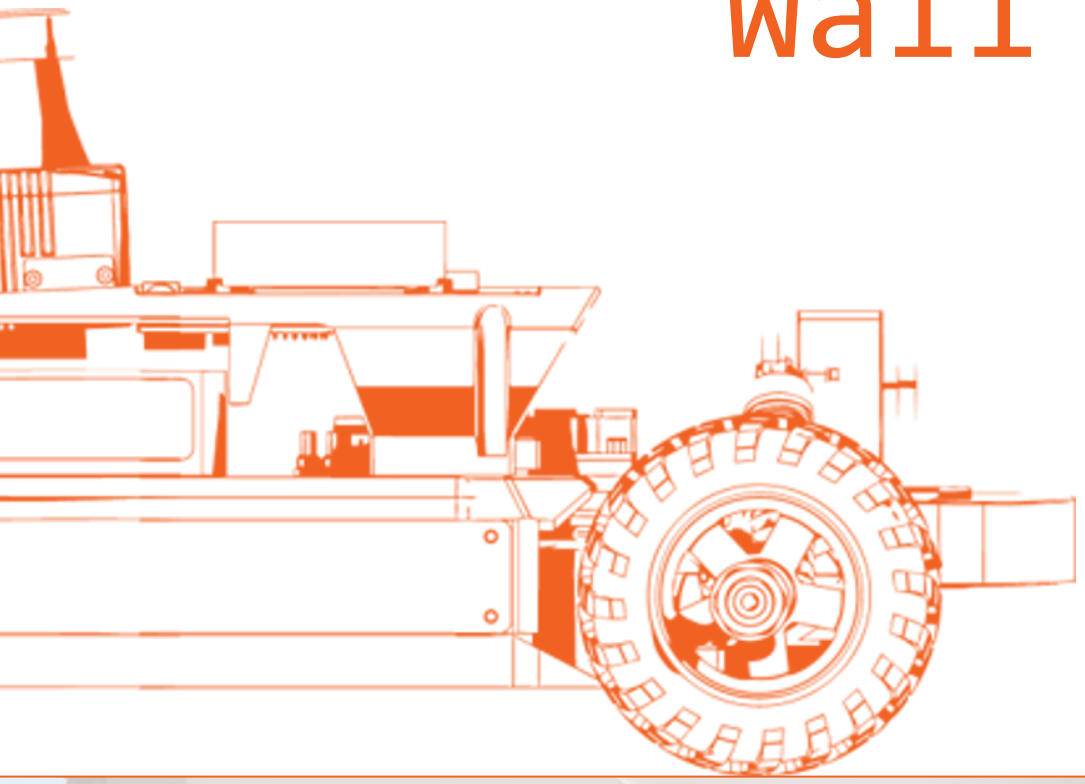
Assessment:
Collision count = 0

Secondary Objective:
Performance

Assessment:
Distance to collision ≤ 1 m

Reference Code

Wall Following



Formulation A: Follow Just One Wall

Let the distance to the right wall at the current time t be D_t

Let the vehicle be oriented at an angle α w.r.t. right wall

Consider two laser beams:

- b exactly $90°$ to the right of the vehicle
- a any arbitrary angle $0° < \theta \leq 70°$ w.r.t. beam b

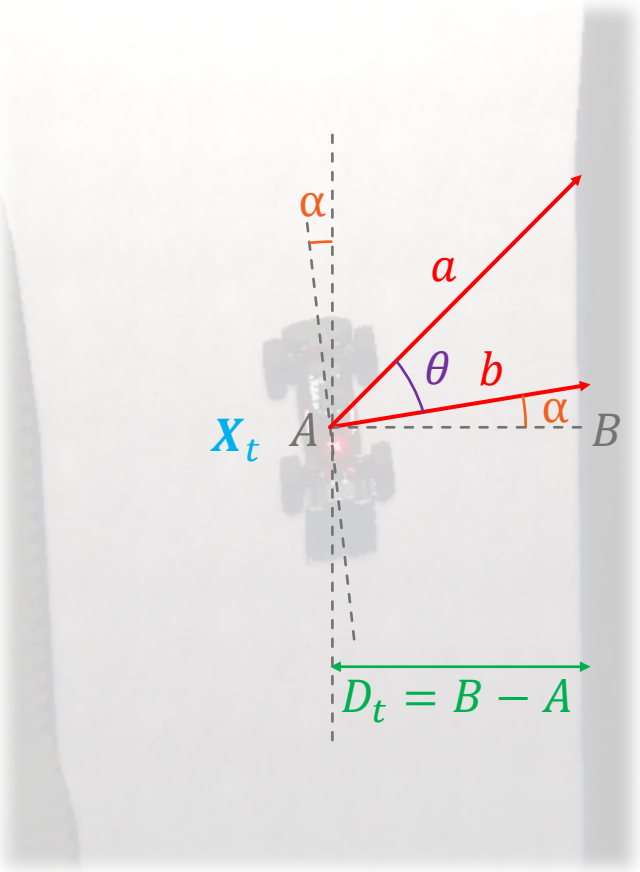
Using trigonometry, we can express $\alpha = \tan^{-1}\left(\frac{a \cdot \cos(\theta) - b}{a \cdot \sin(\theta)}\right)$

We can then express $D_t = b \cdot \cos(\alpha)$

Cross-track error is the difference between D_t and a reference D_r

$$CTE_t = D_r - D_t$$

The lateral controller then acts on this error: $\delta_t = \text{PID}(CTE_t)$



Formulation A: Follow Just One Wall

Small issue: The vehicle will be traveling at a high speeds and will have a non-instantaneous response to whatever speed and steering commands are computed. If we simply use the current distance to the wall, the vehicle might end up turning too late and may crash!

Proposed solution: We can account for this delay and project the vehicle ahead by a certain lookahead distance (let's call it L).

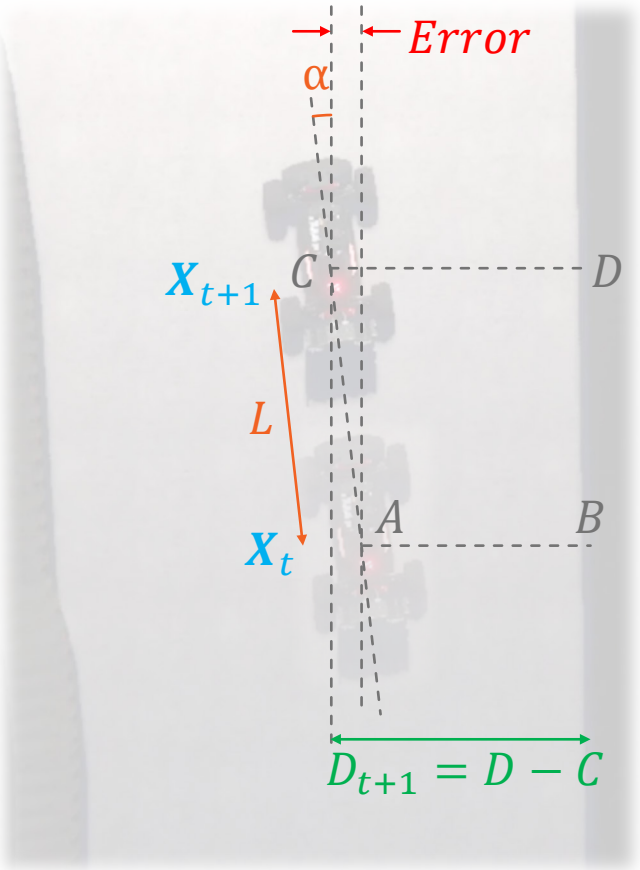
We can express projected distance to the wall $D_{t+1} = D_t + L \cdot \sin(\alpha)$

Cross-track error is now the difference between D_{t+1} and D_r

$$CTE_t = D_r - D_{t+1}$$

The lateral controller then acts on this error: $\delta_t = \text{PID}(CTE_t)$

The longitudinal controller can track a velocity setpoint v_r by controlling the throttle: $\tau_t = \text{PID}(v_r - v)$, or adapt the velocity based on steering.



Formulation B: Follow Both the Walls

Consider two laser beams:

a exactly 90° to the left of the vehicle

b exactly 90° to the right of the vehicle

Let the distance to the left wall at the current time t be a

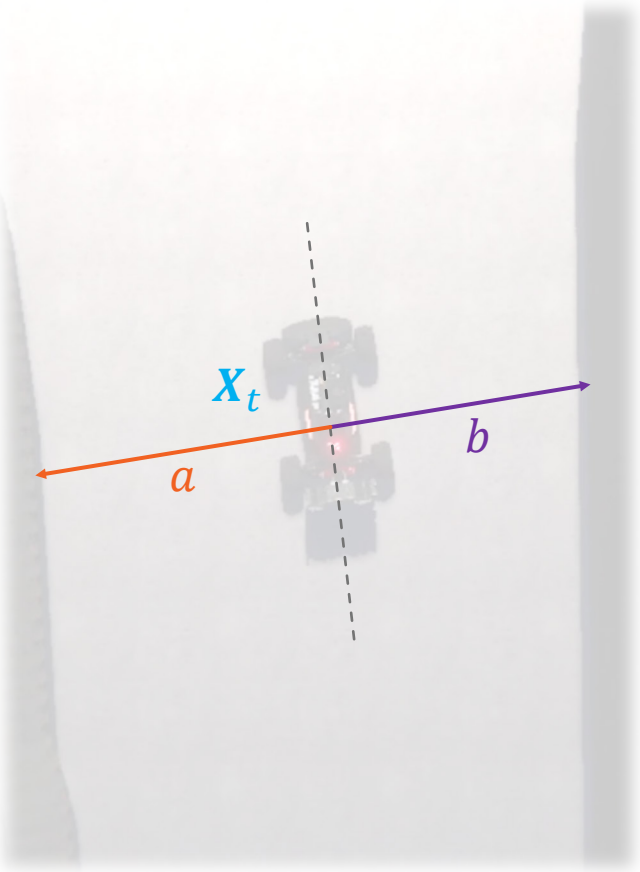
Let the distance to the right wall at the current time t be b

Cross-track error is the difference between a and b

$$CTE_t = a - b$$

The lateral controller then acts on this error: $\delta_t = \text{PID}(CTE_t)$

The longitudinal controller can track a velocity setpoint v_r by controlling the throttle: $\tau_t = \text{PID}(v_r - v)$, or adapt the velocity based on steering.



Formulation B: Follow Both the Walls

Small issue: Using a single measurement can be suboptimal. A spurious return due to any reason (noise, dust, a reflection) can destabilize the controller and compromise its behavior. Vehicle traveling at high speed may not have enough time to recover and may crash!

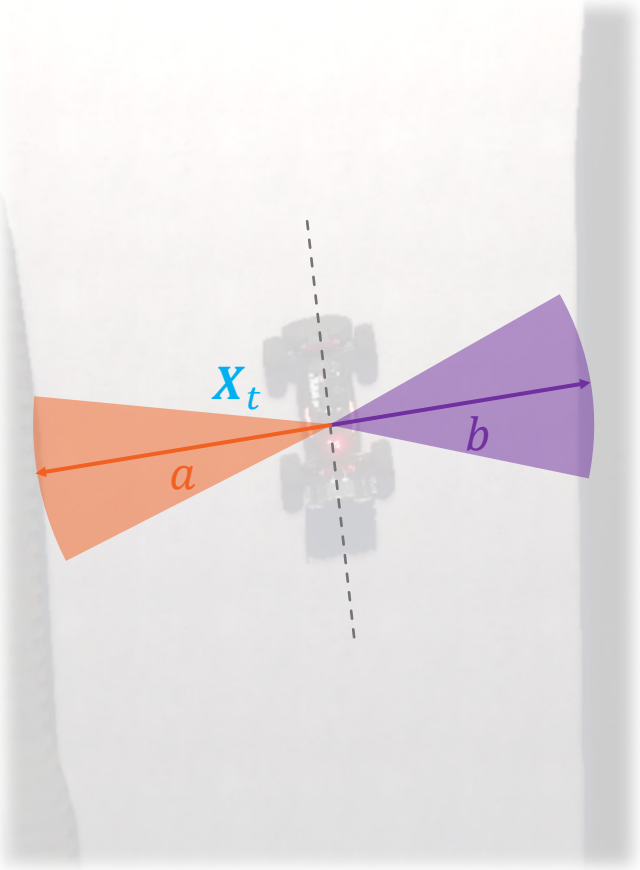
Proposed solution: We can consider multiple range measurements across a small sector on either sides. The left and right distances can be then computed as the mean of these readings, which is more stable/reliable.

Cross-track error is the difference between a and b

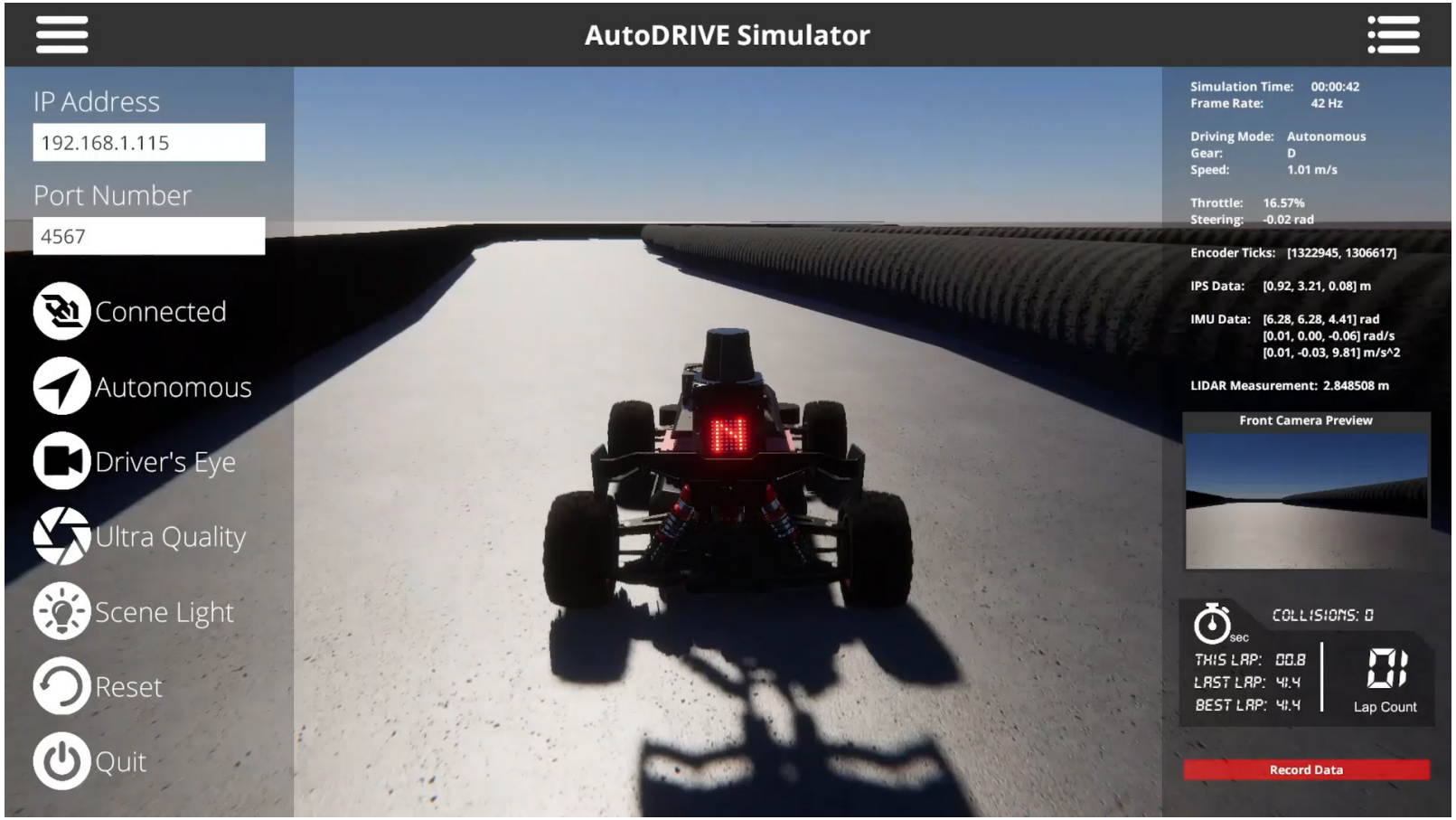
$$CTE_t = a - b$$

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Wall Following Demo



Primary Objective:
Safety

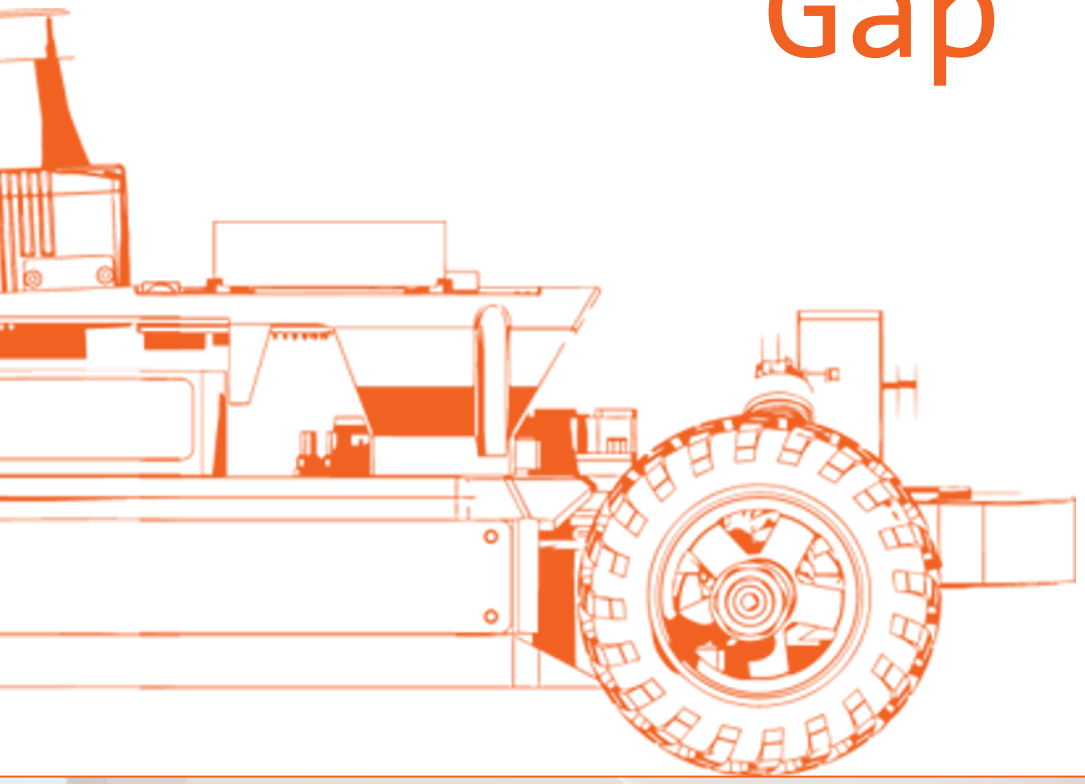
Assessment:
Collision count = 0

Secondary Objective:
Performance

Assessment:
Minimize lap time

Reference Code

Gap Following



Formulation A: Naïve Gap Follower

Step 1: Register LIDAR range array

ranges = […, 2.1, 2.2, 4.8, 4.2, …]

Step 2: Find the angle corresponding to the largest range value

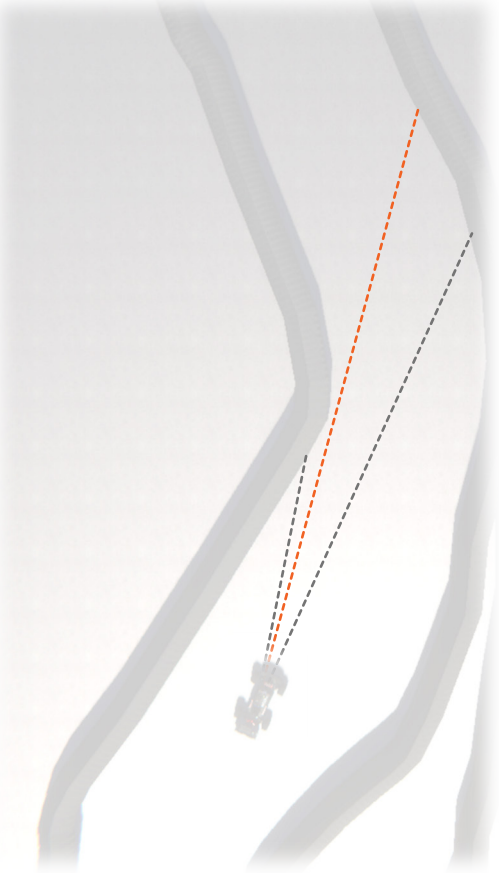
$\underset{\theta}{\text{argmax}}\, \text{ranges}$

Step 3: Drive in that direction

$\delta_t \leftarrow \theta_t$

The longitudinal controller can track a velocity setpoint v_r by controlling the throttle: $\tau_t = \text{PID}(v_r - v)$, or adapt the velocity based on frontal LIDAR measurement(s).

Small issue: The LIDAR rays used to identify the furthest visible point are practically zero-width, but the vehicle has a non-zero width. If the vehicle simply drives towards this point, it might collide with the track bounds!



Formulation A: Naïve Gap Follower

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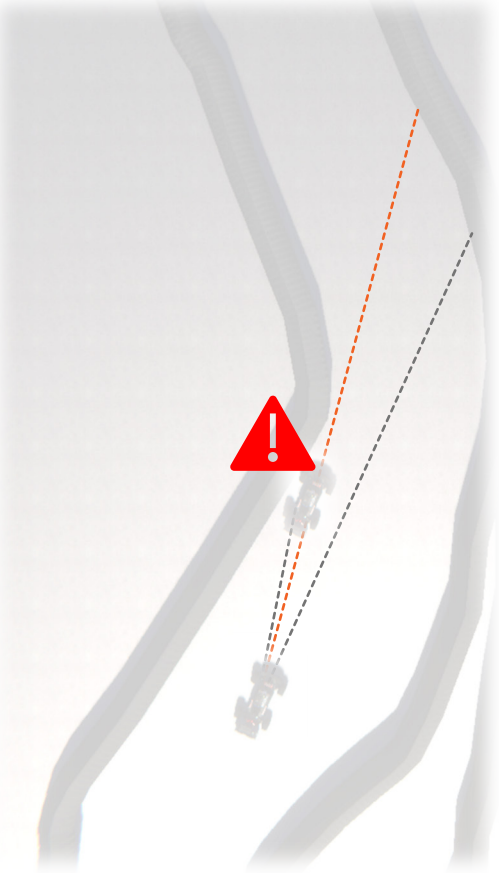
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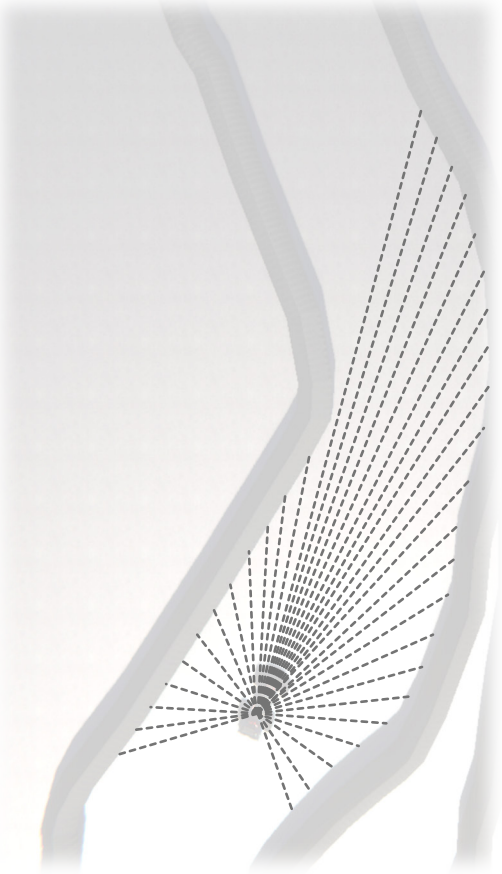
Formulation B: Disparity Extender

Proposed solution: We can account for the vehicle footprint by extending the disparity of LIDAR measurements.

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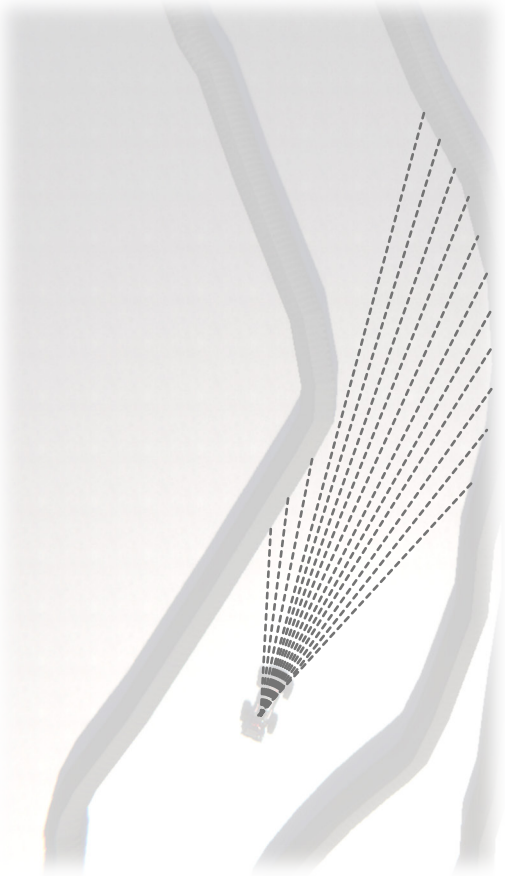


Formulation B: Disparity Extender

Proposed solution: We can account for the vehicle footprint by extending the disparity of LIDAR measurements.

Step 1: Register LIDAR range array […, 2.1, 2.2, 4.8, 4.2, …]

Step 2: Filter frontal measurements […, 2.1, 2.2, 4.8, 4.2, …]



Formulation B: Disparity Extender

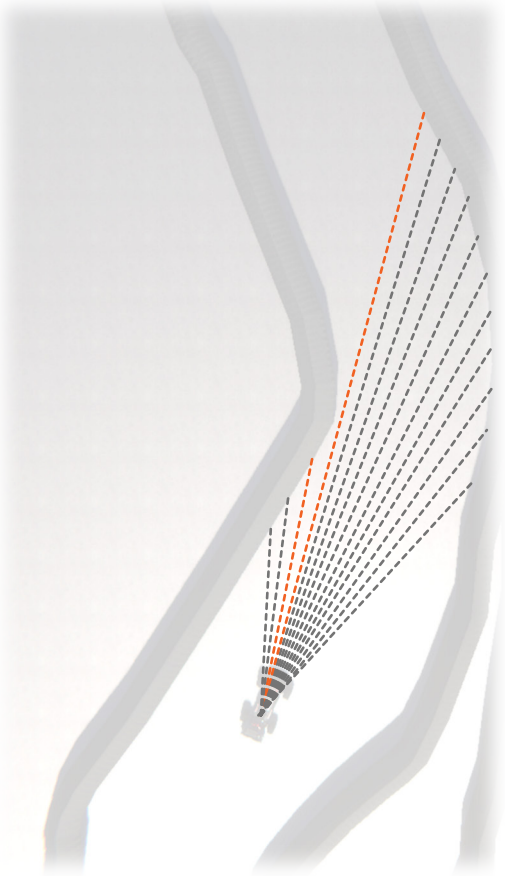
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Step 1: Register LIDAR range array […, 2.1, 2.2, 4.8, 4.2, …]

Step 2: Filter frontal measurements […, 2.1, 2.2, 4.8, 4.2, …]

Step 3: Find a disparity […, 2.1, 2.2, 4.8, 4.2, …]

$$\begin{cases} \text{ranges[i]} - \text{ranges[i+1]} \leq d_{th}; & \text{nominal} \\ \text{ranges[i]} - \text{ranges[i+1]} > d_{th}; & \text{disparity} \end{cases}$$



Formulation B: Disparity Extender

Proposed solution: We can account for the vehicle footprint by extending the disparity of LIDAR measurements.

Step 1: Register LIDAR range array […, 2.1, 2.2, 4.8, 4.2, …]

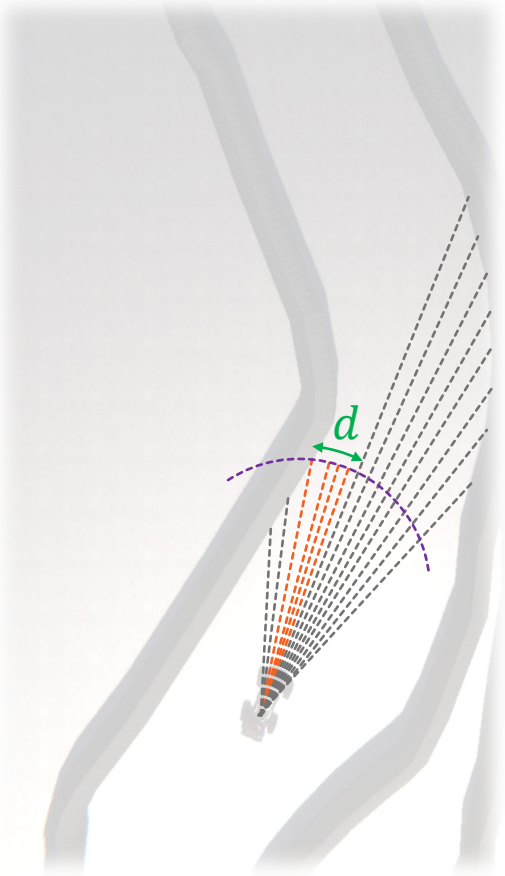
Step 2: Filter frontal measurements […, 2.1, 2.2, 4.8, 4.2, …]

Step 3: Find a disparity […, 2.1, 2.2, 4.8, 4.2, …]

Step 4: Extend the disparity […, 2.1, 2.2, 2.2, 2.2, …]

$$d = \frac{\tan^{-1}\left(\frac{\frac{W}{2} + M}{R}\right)}{\text{LIDAR_ANG_RES}}$$

W: Width of the vehicle footprint
M: Safety margin
R: Range at the closer index



Formulation B: Disparity Extender

Proposed solution: We can account for the vehicle footprint by extending the disparity of LIDAR measurements.

Step 1: Register LIDAR range array […, 2.1, 2.2, 4.8, 4.2, …]

Step 2: Filter frontal measurements […, 2.1, 2.2, 4.8, 4.2, …]

Step 3: Find a disparity […, 2.1, 2.2, 4.8, 4.2, …]

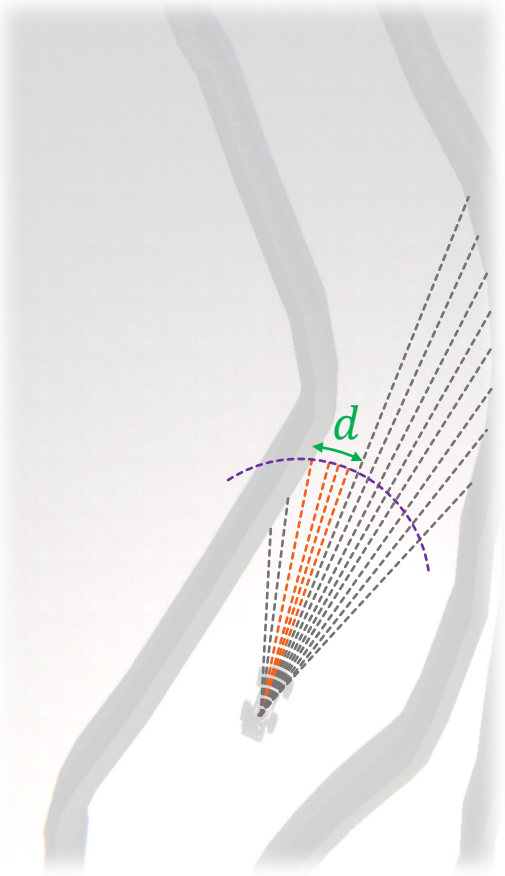
Step 4: Extend the disparity […, 2.1, 2.2, 2.2, 2.2, …]

Repeat this process for every disparity but never overwrite closer ranges with farther ones.

$$\begin{cases} \text{ranges[i]} - \text{ranges[i+1]} \leq d_{th}; & \text{nominal} \\ \text{ranges[i]} - \text{ranges[i+1]} > d_{th}; & \text{disparity} \end{cases}$$

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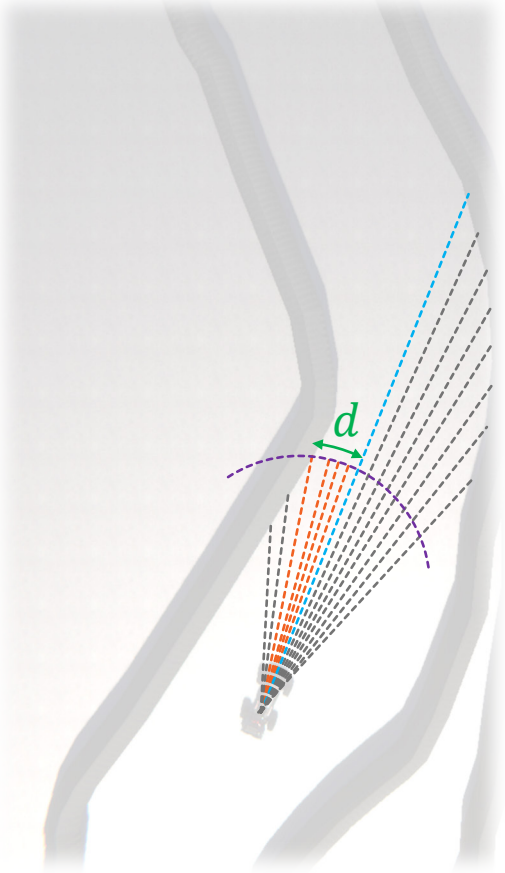
Step 2: Filter frontal measurements […, 2.1, 2.2, 4.8, 4.2, …]

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Step 4: Extend the disparity […, 2.1, 2.2, 2.2, 2.2, …]

Step 5: Target the farthest reachable point

$$\delta_t \leftarrow \theta_t$$



Formulation B: Disparity Extender

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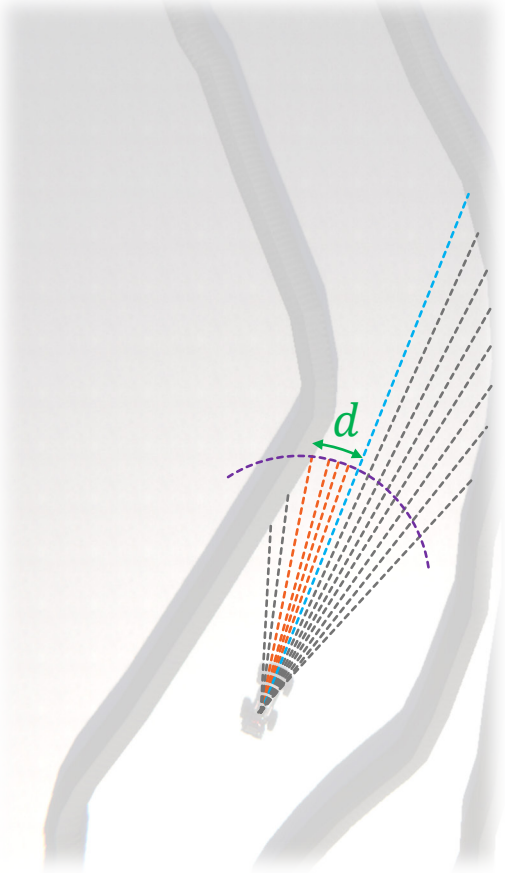
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You can smoothen the steering rate by weighing it against the previous steering command: $\delta_t \leftarrow k_S \cdot \delta_{t-1} + (1 - k_S) \cdot \theta_t$



Formulation B: Disparity Extender

Proposed solution: We can account for the vehicle footprint by extending the disparity of LIDAR measurements.

Step 1: Register LIDAR range array […, 2.1, 2.2, 4.8, 4.2, …]

Step 2: Filter frontal measurements […, 2.1, 2.2, 4.8, 4.2, …]

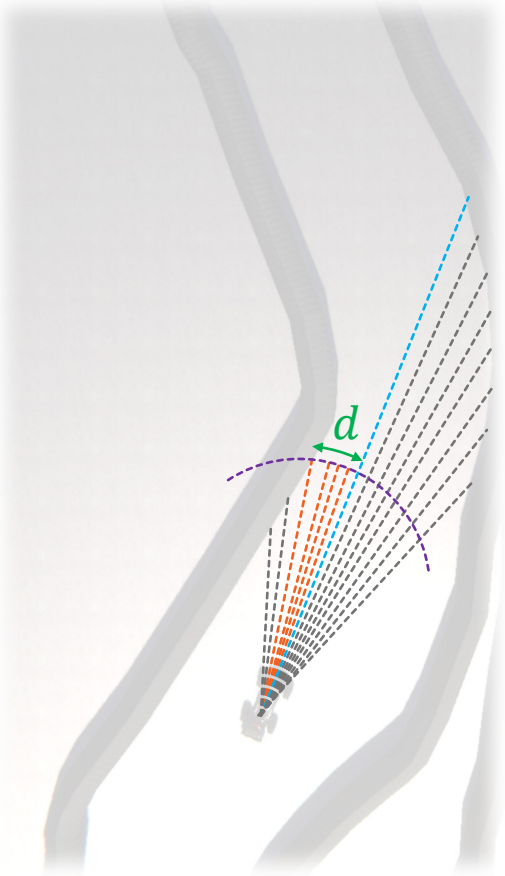
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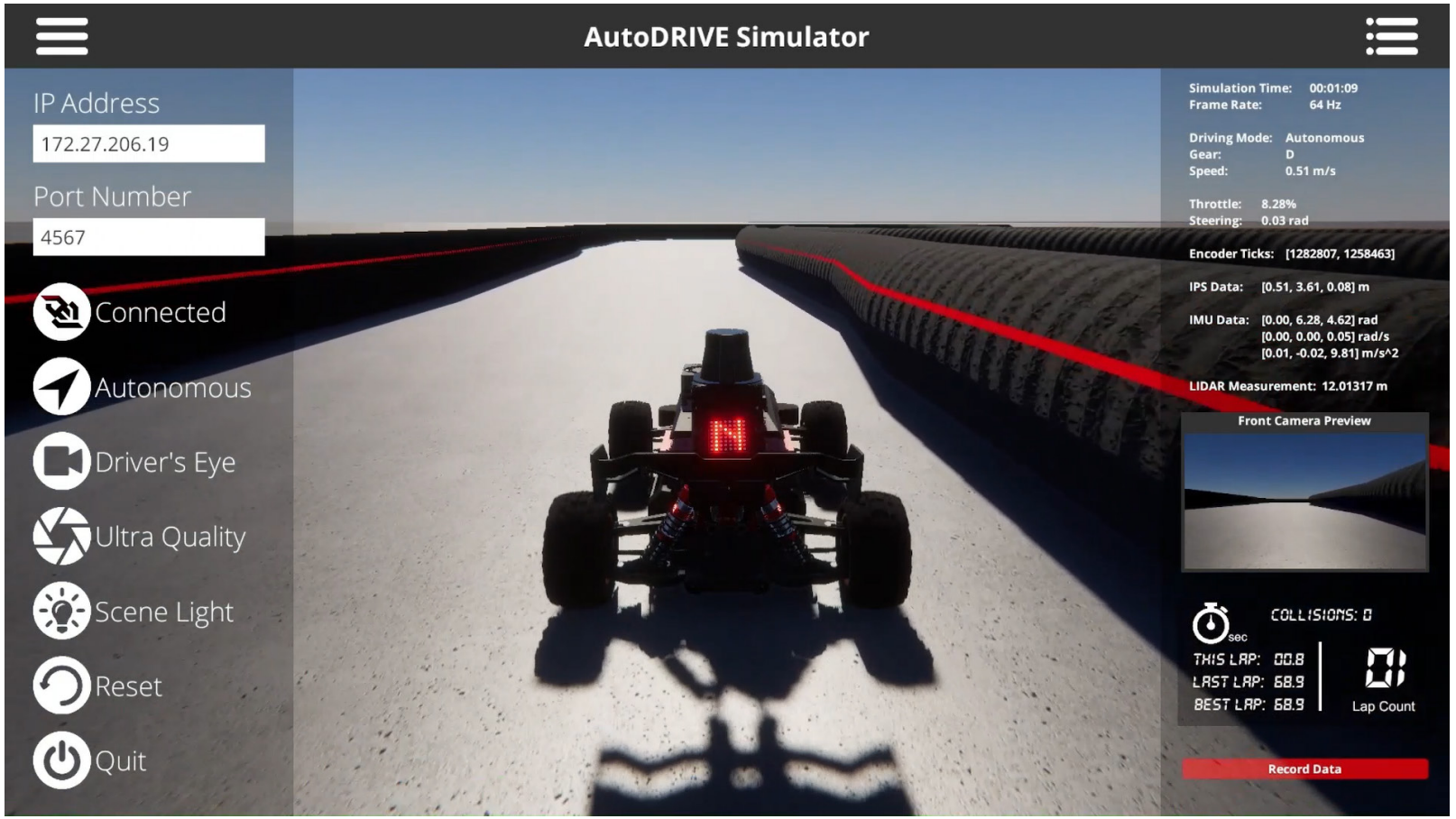
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The longitudinal controller can track a velocity setpoint v_r by controlling the throttle: $\tau_t = \text{PID}(v_r - v)$, or adapt the velocity based on frontal LIDAR measurement(s).



Gap Following Demo



Primary Objective:
Safety

Assessment:
Collision count = 0

Secondary Objective:
Performance

Assessment:
Minimize lap time

Reference Code

Thank You!

Open to questions and discussion…

